Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/13/2005
PI Name:	Lee, Te-Won Ph.D.		
Project Title:	Automatic Speech Recognition in Noisy Environments - Human Factors Approach		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	03/10/2003	**End Date:**	04/10/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	6
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Lee		
COI Name (Institution):			
Grant/Contract No.:	NAG9-1479		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The goal of the second year was the evaluation and testing of the ICA technology developed by SoftMax, Inc. for speech enhancement in noisy shuttle (ISS) environment. To achieve the task goals, we have chosen a commercial recognition system where its recognition rate has been further enhanced by our ICA technology. We have tested the proposed cofiguration with human subjects at JSC on the Robonaut task. In ISS voice recognition capability enhances the human performance. Accurate and reliable recognition is necessary to deploy the system. Our technology yields significant improvements over the existing technology.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The focus of the application has been on Robonaut Command and Control via voice recognition. Test results have shown significant increase in voice recognition accuracy using the ICA technology.
Task Progress:	This document is a progress report on the research performed by SoftMax, Inc., under the NASA/Johnson Space Center (JSC) Grant No. NAG9-1479 for the one-year period March 1, 2004 through February 28, 2005. The title of the project is, "Automatic Speech Recognition in Noisy Environments – A Human Factors Approach," and is a NASA Space Human Factors Engineering project. The principal investigator is Dr. Te-Won Lee, Co-Founder, Chief Executive Officer, and Chief Technical Officer at SoftMax, Inc. Dr. Erik Visser, Senior Research Engineer at SoftMax, was responsible for executing the scientific and technical research and development efforts pertaining to the scope of this project. The purpose of the project is to develop a speech enhancement system based on Independent Component Analysis (ICA), and in particular based on SoftMax, Inc.'s proprietary technology, SoftMax Signal Separation (S^3), for accurate voice recognition in noisy space environments, including the Space Station and the Space Shuttle. An example of a goal is the ability to achieve enhanced and robust voice recognition for voice control of NASA's Robonaut humanoid space robot, particularly in the performance of extravehicular activity (EVA) assembly and maintenance tasks on the Space Station, the Space Shuttle, and potentially other space vehicles and objects (e.g., orbital satellites), as well as, possibly in the future, planetary surface exploration. Another goal is the ability to achieve robust voice recognition under conditions of weightlessness (microgravity), as simulated on the KC-135 parabolic aircraft. To take the Robonaut example further, the Robonaut humanoid space robot is a system currently under design by NASA/JSC that is intended to take the place of human astronauts in rather risky and dangerous spacewalks (EVAs). These EVAs are typically required in order to perform maintenance and repairs on outside components of space vehicles such as the Space Station and the Space Shuttle. Robonaut is modeled on the human body with an upper torso, a moving head with built-in cameras for eyes, and functional arms and hands that can perform grasping, twisting, and other mechanical operations that are typical of human activities when performing mechanical maintenance tasks, such as fastening or unfastening (tightening or removing) bolts. Some of the dangers to humans of these spacewalks are the extreme conditions of space (pressure, temperature, microgravity or weightlessness), the possibility of equipment malfunction, and physiological effects of these conditions on the astronaut, among other things. Far better to have a robot perform outside-vehicle maintenance in order to spare the astronaut of the potential dangers of these activities. One way of operating the Robonaut is through telepresence: a human "teleoperator" communicates with the robot by virtual reality through a donned special helmet and gloves. Movements of the teleoperator's head are mimicked by Robonaut, as are movements of the teleoperator's hands and arms. The teleoperator can simulate tightening a bolt, and Robonaut will mimic this motion to actually tighten a bolt on a piece of equipment. Another aspect of commanding Robonaut is through voice control by the teleoperator. The teleoperator issues a voice command, and Robonaut interprets that command and then takes action based on its interpretation. One way to attempt to achieve robust command recognition in this way, so that the teleoperator's voice commands can be accurately understood and interpreted by Robonaut (so that Robonaut can perform the correct action and not some other action that was undesired or unintended), is by the integration of Automatic Speech Recognition (ASR) software systems into Robonaut's control software. There are a number of leading edge commercial off-the-shelf (COTS) ASR solutions on the market today. However, there are two major problems with traditional ASR systems: the first is that normal human speech communications tend to suffer from a lack of clarity, just because human beings are so used to interpreting each other and "filling in the gaps" of missing information, that even mumbled speech can at times be accurately interpreted by humans, whereas ASR systems will consistently fail to recognize such speech. The second problem with modern ASR systems is that it is often necessary to train the ASR system to the operator's voice, so that it becomes familiar with the person's inflections, accent (national and regional), idiosyncratic ways of pronouncing certain words (e.g., nookyular instead of nuclear), and other personal characteristics, enabling the ASR system to accurately recognize and interpret the person's words and commands. This operator training can be very time-consuming, and isn't always practical in a real-time environment, or in, for example, an emergency. But the biggest monkey wrench in the works of modern ASR systems is the presence of noise. And space vehicles are very noisy environments. The Space Station is noisy. The Space Shuttle is noisy. The whir of engines, steady and intermittent operational noises in the environment, other people talking — sometimes shouting — all of these add stock into the soup of sounds that go into a microphone when a teleoperator is trying to communicate with Robonaut. And if the sound blanches out the teleoperator's spoken commands, then Robonaut has no idea what it is being asked to do. The research being performed by SoftMax has the goal of determining to what extent, and whether, its proprietary S^3 signal separation software can, when integrated into a top-tier modern ASR system, screen out the varieties of noise in a noisy space environment so that, for example, the ASR system can robustly and accurately recognize and interpret the teleoperator's spoken voice commands, enabling Robonaut to perform exactly what it is being asked to do. Part of the study is to test the effectiveness of the "bald" ASR system, absent the presence of S^3, and then to test the ASR system when enhanced by the integration of S^3. In addition to the "what" there is the "how": we test how S^3 can be integrated into the ASR system, and the particular configuration of hardware and software that optimizes the performance of the system and thereby optimizes the success of remote operation of Robonaut by voice command, among other applications. Until the limitations of noisy space environments can be overcome, voice operation of Robonaut will fail to reach its potential, as will other applications of voice recognition in space. It is our hope at SoftMax that these studies yield a system that, among other things, can make voice control of Robonaut not only feasible but cost-effective, and most importantly, successful. It is the stuff of science fiction brought to reality.
Bibliography Type:	Description: (Last Updated: 06/13/2005)

Abstracts for Journals and Proceedings	Lee, T.-W. "Automatic Speech Recognition in Noisy Environments - Human Factors Approach" Automatic speech recognition system will facilitate the effectiveness of the human system interface by providing hands-free command and control functionality for time and safety critical operations such as robotics operations during ISS assembly, shuttle remote manipulator system/closed circuit television system operations, and protein crystal growth experiments or dissection tasks using the glove boxes. The goal of this proposal is to optimize automatic voice recognition systems such that they work accurately in noisy space flight environments. The new technology is based on blind signal separation that allows non-stationary noise to be removed from the target speech signal. Studies and industrial applications have demonstrated that this technology can boost recognition accuracy from 30% to 80% in very noisy environments. Bioastronautics 2005 , Jan-2005